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Interference between Concurrent Resistance and Endurance Exercise: Molecular Bases and the Role of Individual Training Variables

Jackson J. Fyfe · [David J. Bishop](#) · Nigel K. Stepto

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Abstract Concurrent training is defined as simultaneously incorporating both resistance and endurance exercise within a periodized training regime. Despite the potential additive benefits of combining these divergent exercise modes with regards to disease prevention and athletic performance, current evidence suggests that this approach may attenuate gains in muscle mass, strength, and power compared with undertaking resistance training alone. This has been variously described as the interference effect or concurrent training effect. In recent years, understanding of the molecular mechanisms mediating training adaptation in skeletal muscle has emerged and provided potential mechanistic insight into the concurrent training effect. Although it appears that various molecular signaling responses induced in skeletal muscle by endurance exercise can inhibit pathways regulating protein synthesis and stimulate protein breakdown, human studies to date have not observed such molecular ‘interference’ following acute concurrent exercise that might explain compromised muscle hypertrophy following concurrent training. However, given the multitude of potential concurrent training variables and the limitations of existing evidence, the potential roles of individual training variables in acute and chronic interference are not fully elucidated. The present review explores current evidence for the molecular basis of the specificity of training adaptation and the concurrent

interference phenomenon. Additionally, insights provided by molecular and performance-based concurrent training studies regarding the role of individual training variables (i.e., within-session exercise order, between-mode recovery, endurance training volume, intensity, and modality) in the concurrent interference effect are discussed, along with the limitations of our current understanding of this complex paradigm.

1 Introduction

Skeletal muscle is a highly malleable tissue capable of significant metabolic and morphological adaptations in response to disruptions in cellular homeostasis induced by exercise [1, 2]. Resistance and endurance training represent divergent exercise modes, with each inducing distinct responses within the muscle milieu that act to minimize cellular stress during subsequent exercise bouts [3, 4]. In this regard, the skeletal muscle adaptations associated with exercise training are highly specific to the mode of exercise performed (i.e., resistance vs. endurance exercise), along with the frequency, intensity, and duration of the exercise stimulus [5]. For example, chronic resistance training promotes enhanced muscle activation and fiber hypertrophy, resulting in increased maximal contractile force [6, 7]. Conversely, endurance training increases mitochondrial density and oxidative capacity of the trained muscle fibers [8], whilst promoting alterations in substrate metabolism [9], culminating in increased whole-body aerobic capacity (VO_{2max}) [5].

Concurrent training can be defined as the simultaneous integration of resistance and endurance exercise into a periodized training regime. Despite the potentially wide-ranging benefits of combining resistance and endurance

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exercise, there is considerable evidence that concurrent training compromises the development of muscle mass, strength, and power compared with undertaking resistance exercise alone [10–12]. While the mechanisms of the interference effect are likely to be multifactorial, presumably endurance exercise either interferes with the ‘quality’ of resistance exercise sessions (via residual fatigue and/or substrate depletion) [11], and/or compromises the acute molecular responses activated by resistance exercise that mediate fiber hypertrophy [13, 14].

Insight into the molecular factors that mediate the specific adaptations to divergent exercise stimuli has emerged in recent years. Training adaptations in skeletal muscle are considered the cumulative result of acute signaling responses and subsequent gene expression initiated after repeated exercise bouts, resulting in the accumulation of specific proteins over time and, subsequently, an altered muscle phenotype [1, 15, 16]. Although these processes are incompletely resolved, the mammalian (mechanistic) target of rapamycin complex 1 (mTORC1) has been identified as a key mediator of resistance exercise-induced increases in protein synthesis and subsequently muscle growth [17, 18], whereas the 5' adenosine monophosphate (AMP)-activated protein kinase (AMPK) and Ca^{2+} /calmodulin-dependent kinase II (CaMKII) cascades, among others, are activated by endurance exercise and converge on the peroxisome proliferator-activated receptor- γ coactivator-1 (PGC-1 α) to coordinate mitochondrial biogenesis [19, 20]. Of particular relevance to the concurrent interference effect is that various signaling responses activated by endurance exercise appear to inhibit those that regulate muscle hypertrophy [21–23]. Various studies have attempted to detect such ‘molecular interference’ occurring following concurrent exercise in humans [24–30]; however, to date, these studies have failed to replicate an acute interference mechanism as distinct as that observed in rodents [23]. It is therefore unclear whether this mechanism operates in humans, or, rather, our understanding is hindered by the limitations of existing evidence. In addition, given the multitude of concurrent training variables, the potential roles of specific training variables in the interference effect are incompletely resolved. As a result, there is substantial difficulty in deducing practical recommendations from current evidence to simultaneously maximize the benefits of resistance and endurance training.

This review summarizes current evidence for the molecular basis underlying the specificity of training adaptation and the concurrent interference effect. Additionally, the potential role of specific concurrent training variables in the interference effect is discussed, along with the limitations of our current understanding of this complex paradigm.

2 Literature Search

The articles selected for review were obtained via searches of MEDLINE and SPORTDiscusTM between 1957 and September 2013. The following keywords were searched in combination: ‘concurrent training,’ ‘molecular,’ ‘interference,’ ‘exercise,’ and ‘training adaptation.’ From the abstracts returned, articles were included for review if they related to the molecular basis for the specificity of training adaptation, or interference associated with concurrent versus single-mode training. Literature cited in each article chosen was also searched, and additional articles satisfying the above criteria were likewise included for review.

3 Concurrent Training and the Interference Effect

The concomitant integration of resistance and endurance training within a periodized training regime is termed ‘concurrent training’ [11, 12]. Given the capacity of both exercise modes to induce adaptations within skeletal muscle that counteract a number of disorders impacting upon functional capacity and metabolic health, including sarcopenia [31–33], type II diabetes, and obesity [34–36], concurrent training appears to be an attractive exercise strategy for preventing and counteracting multiple disease states. Additionally, from an athletic perspective, concurrent training is necessary for sports whereby improved strength, power, and/or hypertrophy are desired concomitant with enhanced aerobic capacity [37].

Despite the obvious potential benefits of combining resistance and endurance exercise, since the work of Hickson [10] it has been well established that concurrent training often results in compromised adaptation compared with training for either exercise mode alone [11]. This phenomenon has been variously described as the interference effect or concurrent training effect [12, 14]. The interference effect typically manifests as a compromised resistance training adaptation compared with undertaking resistance training alone. For example, previous research demonstrates that concurrent training, relative to resistance training only, results in compromised strength [10, 38–41], hypertrophy [10, 41, 42], and power development [40, 41, 43–45]. Conversely, resistance training appears to have little to no negative impact on endurance performance and $\text{VO}_{2\text{max}}$ [11, 12], although compromised gains in aerobic capacity with concurrent versus endurance training alone have been reported [44, 46]. However, concurrent training can augment both short- (<15 min) and long-duration (>30 min) endurance performance, predominantly via improvements in neuromuscular function and economy [47, 48].

4 Molecular Basis for Training Adaptation Specificity

Resistance and endurance exercise represent divergent exercise modes, both with regards to their inherent stimuli and the subsequent adaptations induced within skeletal muscle by chronic training. In recent years, insight into the molecular factors that mediate skeletal muscle adaptations to divergent exercise modes has emerged [17, 23, 49, 50]. Exercise-induced adaptations in skeletal muscle are considered the cumulative result of acute molecular signaling responses and subsequent gene expression initiated after repeated exercise bouts, leading eventually to the accumulation of specific proteins over time and, subsequently, an altered muscle phenotype [1, 15, 16]. It has been suggested that resistance and endurance exercise stimulate almost distinct activation of specific molecular signaling pathways and gene networks that mediate the mode-specific adaptations to chronic exercise training [2, 14, 23].

The principal adaptation to chronic resistance exercise is perhaps muscle fiber hypertrophy [51], which is the cumulative result of transient increases in net protein synthesis above that of protein breakdown occurring for up to 48 h post-exercise [52, 53]. It is generally accepted that fiber hypertrophy consequent to resistance exercise is mediated by the anabolic mTORC1 signaling cascade [17]. The mTORC1 pathway integrates signals from mechanical stimuli, growth factors, and nutrients to promote increased net protein synthesis by phosphorylating downstream targets implicated in translation initiation (i.e., p70S6K [70 kDa ribosomal protein S6 kinase 1] and 4E-BP1 [eIF4E binding protein 1]) [17, 54]. The pivotal role for mTORC1 in load-induced muscle growth is supported by evidence that rapamycin (a selective mTORC1 inhibitor) administration prevents both muscle hypertrophy *in vivo* [17] and the increase in muscle protein synthesis following acute resistance exercise in humans [18].

In contrast to resistance exercise, endurance exercise is typically characterized by lower intensity, longer-duration contractile activity that imparts far less mechanical strain on the active muscle fibers [55]. This presents a significant metabolic challenge within the muscle milieu, resulting in perturbations in intracellular concentrations of Ca^{2+} , oxygen, lactate, reactive oxygen species (ROS), and increased AMP:ATP (adenosine triphosphate) and $\text{NAD}^+:\text{NADH}$ (nicotinamide adenine dinucleotide: NAD^+ reduced form) ratios [2]. These stimuli then initiate a number of intracellular signaling cascades, including the 5' AMPK and $\text{Ca}^{2+}/\text{CaMKII}$ pathways, which converge on PGC-1 α to promote mitochondrial biogenesis [19, 49, 56], and other associated adaptations, including improved substrate utilization [9] and capillary density [57], which collectively enhance oxidative capacity [5].

Early insight into the molecular basis for the specificity of training adaptation came from the work of Atherton and colleagues [23]. These workers employed a model in which isolated rat muscle was electrically stimulated at either high (six sets of ten 3-s repetitions at 100 Hz for 20 min) or low (3 h at 10 Hz) frequencies to mimic resistance or endurance exercise, respectively. Their results indicated selective activation of the anabolic Akt/mTOR signaling cascade by resistance-like stimulation with little effect on the AMPK/PGC-1 α pathway, while endurance-like stimulation caused increased AMPK activation and PGC-1 α protein levels [23]. Moreover, endurance-like stimulation inhibited Akt/mTOR and its downstream targets. Therefore, these authors postulated that selective activation of either Akt/mTOR or AMPK/PGC-1 α could explain the divergent adaptations associated with resistance and endurance training, respectively, in a paradigm termed the 'AMPK/Akt master switch' hypothesis [23]. Whilst this is an attractive regulatory model, the existence of such clear divergence in humans has, to date, proven elusive [58–62]. This is perhaps not surprising, given that the *ex vivo* electrical stimulation rat model employed by Atherton and colleagues [23] arguably represents few similarities to contracting human skeletal muscle during exercise. Indeed, several observations question the simplistic notion of an AMPK/Akt 'master switch' mediating training adaptation specificity in human skeletal muscle, whilst highlighting the complexity of exercise-induced molecular responses. For example, a number of human studies have failed to detect notable differences in either mTORC1 or AMPK signaling in the early recovery period following endurance and resistance exercise performed separately [58, 61, 62]. Several studies have shown increased mTORC1 activity following endurance exercise in humans [63–65], which may reflect the putative role for mTORC1 in regulating oxidative metabolism [66–68], whilst resistance exercise can acutely activate AMPK [27, 60–62, 69, 70]. Furthermore, mTORC1 can be activated independently of Akt via mechanotransduction [71, 72] and amino acid provision [73, 74], highlighting the discordance between mTORC1 activation mediated via the canonical insulin/insulin-like growth factor (IGF)-1/Akt pathway and contractile activity [75, 76]. Regardless of the exercise modality, other independent factors can modulate the molecular responses to exercise, including training status [59, 60, 63, 77], age [78, 79], genetic factors [80, 81], and nutrient availability [73, 82–84]. The molecular factors mediating the specificity of training adaptation are clearly more complex than dictated by a simplistic 'master switch' model [23], and further work is required to more completely define the mechanisms responsible.

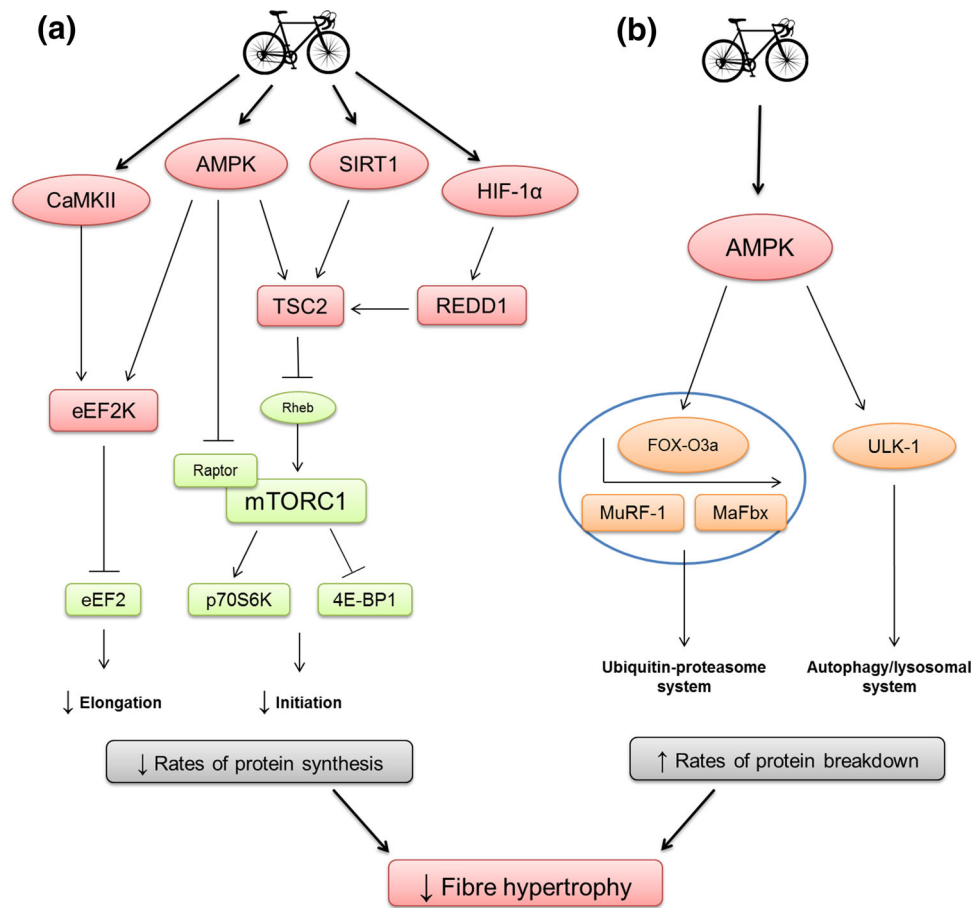


Fig. 1 Putative molecular mechanisms by which endurance exercise potentially **a** inhibits signaling regulating protein synthesis and **b** up-regulates pathways mediating protein degradation, subsequently limiting muscle fiber hypertrophy following concurrent training. AMPK adenosine monophosphate-activated protein kinase, CaMKII calcium/calmodulin-dependent kinase II, eEF2 eukaryotic elongation factor 2, eEF2K eukaryotic elongation factor 2 kinase, FOX-O3a forkhead-box O3a, HIF-1α hypoxia-inducible factor-1α, MaFbx

muscle-atrophy f-box (atrogin 1), mTORC1 mammalian (mechanistic) target of rapamycin complex 1, MuRF-1 muscle ring-finger 1, p70S6K 70 kDa ribosomal s6 protein kinase, REDD1 regulated in DNA development and damage 1, Rheb Ras homologue enriched in brain, SIRT1 sirtuin deacetylase 1, TSC2 tuberous sclerosis complex 2, ULK-1 Unc-51-like kinase 1, 4E-BP1 eIF4E binding protein 1, ↑ indicates increased/greater, ↓ indicates decreased/less

5 Molecular Basis for the Concurrent Interference Effect

Although the molecular signaling mechanisms regulating the specificity of training adaptation are incompletely resolved, there appear to be multiple signaling responses induced by endurance exercise capable of inhibiting protein synthesis and stimulating protein breakdown (Fig. 1). Given that muscle fiber hypertrophy requires a positive net balance of protein synthesis above that of protein breakdown [53], the repeated antagonism of these responses by endurance exercise might contribute to limiting fiber hypertrophy following concurrent training [13, 14, 85].

Perhaps the most well characterized such mechanism involves the putative antagonism between the AMPK and mTORC1 signaling cascades [86–88], thought to be predominantly involved in endurance and resistance training

adaptation [2, 13], respectively. Multiple lines of evidence suggest AMPK activation has a significant inhibitory effect on mTORC1 and its downstream signaling targets, thereby negatively regulating protein synthesis and hypertrophy [21–23, 89–91]. Indeed, AMPK phosphorylation negatively correlates with muscle hypertrophy [92] and is associated with attenuated hypertrophy [93] in rodent functional overload models. Activation of AMPK has been shown to repress mTORC1 signaling via multiple mechanisms in vitro, including direct phosphorylation of the tuberous sclerosis complex 2 (TSC2) [89, 94] and the mTORC1-associated regulatory protein, raptor [91]. Activation of TSC2 by AMPK negatively regulates mTORC1 via inhibition of its upstream activator Rheb (Ras homologue enriched in brain), thereby blocking the downstream activation of regulators of protein translation (i.e., p70S6K and 4E-BP1), and subsequently inhibiting protein synthesis

[21, 69, 89]. This effect is opposed by Akt, which phosphorylates and inactivates TSC2, alleviating its inhibition of mTORC1 [90, 95]. Interestingly, the regulation of mTORC1 by AMPK may be isoform-specific. For example, it appears the AMPK- α 1 catalytic isoform is selectively responsible for limiting muscle hypertrophy via mTORC1 inhibition [87, 94, 96], while AMPK- α 2 governs metabolic adaptations in skeletal muscle [87, 94, 97]. Indeed, AMPK- α 1 is activated following chronic overload in rodents [94], and genetic knockout of this isoform results in greater hypertrophy [96], supporting the isoform-specific role of AMPK in constraining muscle growth. Accumulating in vitro evidence also suggests that, in addition to suppressing protein synthesis, AMPK activation promotes protein degradation via both the ubiquitin-proteasome and autophagy-lysosomal systems [98, 99]. AMPK activation promotes forkhead-box O (FoxO)-dependent transcription of MuFbx (muscle atrophy f-box [atrogen-1]) and MuRF-1 (muscle ring-finger 1) [99–101] and disrupts the inhibitory effect of mTORC1 on ULK1 (Unc-51-like kinase 1) whilst increasing ULK1 activity, leading to autophagy induction [99, 102]. Therefore, the activation of AMPK by endurance exercise potentially mediates interference via down-regulating protein synthesis and concomitantly up-regulating protein degradation [2].

Protein synthesis is also regulated in the elongation phase of protein translation, which is mediated by elongation factors and is the most energy-consuming stage of protein synthesis [103]. The eukaryotic elongation factor 2 (eEF2) is a critical component of the translational machinery involved in translocation of the ribosome along the messenger RNA (mRNA) [104]. The eEF2 is phosphorylated (i.e., inactivated) by eEF2 kinase (eEF2K) [105], which is activated by signaling pathways that respond to increased energy demand or reduced energy supply, such as the AMPK and CaMK pathways, both of which are activated following endurance exercise [106, 107]. Conversely, signaling related to resistance exercise (i.e., mTORC1 and p70S6K activation) inhibits eEF2K activity in vitro, thus releasing its inhibition of eEF2 and increasing translation and protein synthesis rates [108–110]. The activation of eEF2K by endurance exercise is therefore a candidate inhibitor of muscle protein synthesis and, potentially, muscle fiber hypertrophy during concurrent training.

Another upstream inhibitor of mTORC1 and subsequently protein synthesis is REDD1 (regulated in DNA damage and development 1) [111, 112]. REDD1 is activated by a number of metabolic stressors including ATP depletion [111] and hypoxia [113–115], and is induced by endurance exercise in rat skeletal muscle [116]. Upon activation, REDD1 inhibits mTORC1 indirectly by releasing the inhibition of TSC2 caused by 14-3-3 protein binding [114, 115]. Overexpression of REDD1 in rodent skeletal

muscle has been shown to cause a 10 % reduction in muscle fiber size [115], and REDD1 expression is associated with muscle atrophy in diabetic mice [117]. The expression of REDD1 mRNA is reduced 3 h following low-intensity resistance exercise and blood flow restriction in humans, concomitant with increased mTORC1 mRNA expression [118]. Thus, activation of REDD1 by endurance exercise may be an additional mechanism responsible for inhibiting anabolic responses induced by resistance exercise, and subsequently hypertrophy during concurrent training.

The sirtuin (SIRT) deacetylase family of proteins are sensitive to metabolic perturbations, including increased NAD^+ and lactate concentrations, and are activated by endurance exercise in skeletal muscle [119]. Of the SIRT family expressed in skeletal muscle, SIRT1 has been implicated as a potential regulator of mitochondrial biogenesis, in part because it can regulate the activity of AMPK and PGC-1 α [120]. Interestingly, and of potential relevance to concurrent training, is that SIRT1 negatively regulates mTORC1 in vitro [121]. Activated SIRT1 interacts with and subsequently activates TSC2, thereby down-regulating mTORC1 activity [121], potentially via inhibition of the upstream mTORC1-activator Rheb [22]. Increased SIRT1 activity induced by endurance exercise is therefore another potential mechanism by which the mTORC1 pathway and protein synthesis might be suppressed following concurrent exercise.

Collectively, there appears to be multiple signaling responses initiated by endurance exercise with the capacity to inhibit components of the translational machinery and subsequently rates of protein synthesis, in addition to promoting protein breakdown. However, it is important to consider that many of these putative interference mechanisms have been described in cell culture or animal models, and often during non-physiological conditions, which may have limited relevance to human skeletal muscle during exercise [122]. Many of these mechanisms are poorly characterized in skeletal muscle, let alone in response to exercise, and further work is required to confirm their relevance to training adaptations in human skeletal muscle. Indeed, few studies have investigated whether these mechanisms operate in humans following concurrent exercise, and therefore potentially contribute to compromised fiber hypertrophy following concurrent training.

6 Evidence for Molecular Interference in Human Skeletal Muscle

6.1 Molecular Interference Following Acute Concurrent Exercise

Whilst few studies performed to date have investigated divergences between molecular responses to resistance and

Table 1 A summary of current evidence regarding acute molecular interference following concurrent resistance and endurance exercise in humans

Study	Sample size	Participant training status	Study design	Exercise protocols		Results			Conclusions
				Endurance exercise	Resistance exercise	Recovery between modes	Protein phosphorylation	Gene expression	
Carrithers et al. [25]	12 (6 M, 6 F)	1–2 bouts of AE and RE $\times$ /week	Unilateral cross-over. All participants performed unilateral AE then bilateral RE	90 min unilateral cycling at 60 % W_{peak}	Unilateral leg press and leg extension (3 $\times$ 10 reps at 80 % IRM + one set to failure) performed on both legs	30 min	N/A	N/A	Myofibrillar FSR not different between the AE + RE and the RE legs
Coffey et al. [28]	8 M	Regular concurrent training (>1 year)	Randomized cross-over. All participants performed both AE and RE in alternate orders	30 min continuous cycling at 70 % VO_{2peak}	8 $\times$ 5 leg extension reps at 80 % IRM	15 min	$\uparrow$ p-Akt when RE followed AE. No significant order effect was noted for p-TSC2/mTOR/p70S6K	$\uparrow$ MuRF-1 mRNA when AE followed RE. Reverse order resulted in $\downarrow$ IGF-1 mRNA expression	Concurrent exercise does not suppress post-RE myofibrillar protein synthesis rates independent of muscle glycogen levels
Coffey et al. [27]	6 M	Regular concurrent training (>3 $\times$ /week)	Randomized cross-over. All participants performed both AE and RE in alternate orders	10 $\times$ 6-s maximal cycling ergometer sprints against 0.75 Nm torque ⁻¹ passive rest between efforts)	8 $\times$ 5 leg extension reps at 80 % IRM	15 min	Initial RE $\uparrow$ p-p70S6K and p-rps6 but this $\downarrow$ when RE followed repeated sprints. $\uparrow$ p-Akt when RE followed AE. Changes in p-TSC2, p-mTOR, and p-AMPK modest and independent of order	$\uparrow$ MuRF-1 mRNA when sprints followed RE. $\downarrow$ IGF-1 mRNA from rest independent of order	Repeated sprints diminished the anabolic response to RE by attenuating anabolic and enhancing catabolic responses in early recovery

Table 1 continued

Study	Sample size	Participant training status	Study design	Exercise protocols		Results			Conclusions
				Endurance exercise	Resistance exercise	Recovery between modes	Protein phosphorylation	Gene expression	Protein synthesis
Wang et al. [30]	10 (7 M, 3 F)	No programmed exercise for >6 months	Randomized cross-over. All participants performed both AE alone (trial 1) and AE followed by RE (trial 2)	60 min continuous cycling at 65 % VO_{2max} (3-min rest allowed after 30 min)	Six sets of leg press at 70, 75, 80, 80, 75, 70 % 1RM	15 min	↑ p-mTOR and p-p70S6K, ↓ p-eEF2 after AE + RE vs. AE. Similar ↑ in p-Akt, and p-AMPK in AE + RE and AE. No p-CaMKII after each protocol. Heterogenous p-p38 MAPK response	↑ PGC-1α and PDK-4 mRNA with AE + RE vs. AE alone	N/A Addition of RE after an acute AE bout ↑ mitochondrial biogenesis and substrate metabolism signaling and may be a novel method for enhancing aerobic capacity
Lundberg et al. [29]	9 M	AE 2–3×/week and/or RE 1–2×/week for >1 year	Unilateral cross-over. All participants performed unilateral AE followed by bilateral RE 6 h post-AE	40 min continuous unilateral cycling at 70 % of W_{max} . Workload then increased by ~20 W for cycling to exhaustion	Unilateral leg press and leg extension (2 × 7 reps of each exercise, 90-s rest between sets) performed on both legs	6 h	↑ p-mTOR, p-p70S6K, with AE + RE vs. RE. p-rps6 and p-eEF2 unchanged over time for both legs, trend for ↑ p-rps6 with AE + RE	↑ PGC-1α, VEGF, atrogen-1 mRNA with AE + RE vs. RE at PRE and 15 min post. ↓ Myostatin levels in AE + RE vs. RE at PRE and 15 min post. MuRF-1 mRNA similar across legs	N/A Completing RE 6 h after AE did not compromise mTOR-related signaling after leg press and leg extension exercise

Table 1 continued

Study	Sample size	Participant training status	Study design	Exercise protocols		Results		Conclusions		
				Endurance exercise	Resistance exercise	Recovery between modes	Protein phosphorylation			
Donges et al. [26]	8 M	Sedentary middle-aged men (no regular activity involving >30 min/week for 1 year prior to study)	Repeated measures. All participants completed 3 trials in randomized order: (i) RE only, (ii) AE only and (iii) RE + AE combined (50 % volume of each isolated session)	40 min continuous cycling at 55 % of W_{peak}	8 × 8 leg extension at 70 % 1RM	None	↑ p-Akt at 1 h post in CE vs. RE and AE. No change in p-mTOR, p-p70S6K, p-AMPK, p-MAPK, and p-4E-BP1 at any time point. ↑ p-AS160 at 1 and 4 h post for RE. ↑ p-rps6 at 1 h in RE vs. CE and AE	MyoG and MyoD expression ↑ 4 h post RE. MyoG ↑ >AE and CE at 1 h post, and AE at 4 h post. ↑ MyoD mRNA >CE at 1 h post, and AE at 4 h post. No change in myostatin mRNA	Myofibrillar FSR during 4-h recovery ↑ 1.8 and 2.2-fold for RE and CE trials, respectively. ↑ Myofibrillar FSR for CE/RE both significantly >AE, which remained unchanged	CE is as effective as either isolated mode in stimulating myofibrillar and mitochondrial FSR in sedentary middle-aged men despite 50 % less training volume of each modality
Apro et al. [24]	10 M	AE 1–2×/week and RE 2–3×/week for >6 months	Randomized cross-over. All participants performed RE followed by AE, or RE alone	30 min cycling at 70 % VO_{2max}	10 sets of leg press (4 × 8–10 at 85 % 1RM, 4 × 10–12 at 75 % 1RM, 2 sets to fatigue at 65 % 1RM)	15 min	↑ p-mTOR and p-p70S6K and ↓ p-eEF2 regardless of condition. ↓ p-AMPK and p-ACC regardless of condition	↑ PGC-1α mRNA after RE + AE vs. AE	N/A	Endurance exercise performed subsequent to RE does not blunt mTORC1-related signaling

ACC acetyl-CoA carboxylase, AE aerobic exercise, AMPK adenosine monophosphate-activated protein kinase, AS160 Akt-substrate 160 kDa, CaMKII calcium/calmodulin-dependent kinase II, CE concurrent exercise, eEF2 eukaryotic elongation factor 2, F female, FSR fractional synthesis rate, IGF-1 insulin-like growth factor 1, M male, MAPK mitogen-activated protein kinase, mRNA messenger RNA, mTORC1 mammalian (mechanistic) target of rapamycin complex 1, MuRF-1 muscle ring-finger 1, MyoD myogenic differentiation 1, MyoG myogenin, N/A not available, Nm newton-meters, p phosphorylation/phosphorylated, PDK pyruvate dehydrogenase kinase 4, PGC-1α peroxisome proliferator-activated receptor-γ coactivator-1α, PRE pre-resistance exercise, p70S6K 70 kDa ribosomal S6 protein kinase, RE resistance exercise, rept(s) repetition(s), rps6 ribosomal protein S6, TSC2 tuberous sclerosis complex 2, VEGF vascular endothelial growth factor, VO_{2max} maximal oxygen consumption, VO_{2peak} peak oxygen consumption, W watts, W_{peak}/W_{max} peak power output, IRM one repetition maximum, ↑ indicates increased/greater, ↓ indicates decreased/less

endurance exercise in humans [58, 61, 62, 123], more specifically, limited data are available regarding the acute molecular responses following concurrent exercise [25–30] (Table 1). To date, most existing studies have examined molecular responses induced acutely following concurrent exercise cessation (i.e., 15 min to 4 h post-exercise) to investigate whether these responses are altered with concurrent versus single-mode exercise [25–30]. This approach has provided insight into the existence of acute ‘molecular interference’ in humans, and the potential impact of the particular concurrent exercise protocol on chronic training adaptation. However, current evidence is equivocal with regards to the existence of this phenomenon in humans. For example, while some studies have suggested that concurrent training promotes acute interference of anabolic molecular responses [27, 28], others have concluded that either protein synthesis rates [25, 26] or mTORC1 signaling [24] are no different than resistance exercise alone. Remarkably, there is also evidence that concurrent exercise ‘potentiates’ acute adaptive responses to exercise compared with single-mode exercise [30]. For example, the addition of resistance exercise immediately following endurance exercise reportedly augments signaling related to mitochondrial biogenesis [30], whilst performing resistance exercise 6-h after endurance exercise enhances anabolic signaling [29] compared with resistance exercise alone.

Current data therefore provide limited evidence for acute molecular interference with concurrent exercise, and indeed little mechanistic insight into the concurrent training effect. However, as discussed subsequently (see Sect. 7 of this review), the limitations of existing studies must be considered when interpreting evidence with regards to the molecular interference phenomenon in humans. Additionally, given the multitude of variables associated with concurrent training (e.g., resistance/endurance training modality, volume and intensity of exercise, length of between-mode recovery, nutritional and training status of participants), it is difficult to generalize the findings of existing studies outside of the particular design employed. Further work is therefore needed to examine the role of these additional variables in the concurrent interference effect, whilst addressing the limitations of current evidence.

6.2 Molecular Interference Following Short-Term Concurrent Training

Whilst limited acute molecular concurrent training data exist, even less information exists regarding the effects of concurrent training on molecular responses and/or adaptations within skeletal muscle following a period of training [124, 125]. Using a unilateral training model, Lundberg

et al. [124] examined the effect of 5 weeks of concurrent training versus resistance training alone on muscle fiber cross-sectional area (CSA), isokinetic/isometric strength and basal expression of selected regulatory genes (i.e., myostatin, MuRF-1, MaFbx, PGC-1 α , and vascular endothelial growth factor [VEGF]). Interestingly, these authors showed more robust quadriceps femoris hypertrophy when resistance exercise was preceded by aerobic exercise, compared with resistance training alone, although no between-limb difference in isometric strength was noted [124]. No differences in the basal expression of selected genes were observed after training. However, no measures of resting protein content or acute signaling responses to exercise bouts before and after training were obtained, potentially limiting any mechanistic insight into the adaptive outcomes. Similarly, another investigation examined basal phosphorylation and content of selected proteins following 8 weeks of concurrent versus single-mode training [125]. Despite no between-group differences in measures of muscle strength or quadriceps CSA after training, the results showed an increase in basal Akt and AMPK phosphorylation for the resistance- and endurance-only groups, respectively, whilst the concurrent training group showed increased p70S6K protein content [125]. However, given that the exercise protocols employed in these short-term studies were insufficient to cause any interference effect [124, 125], the interpretation of the observed results is difficult. Further work is required to evaluate the effect of long-term concurrent training on anabolic responses and adaptations to concurrent exercise to provide greater insight into the molecular factors that potentially mediate the concurrent training effect.

7 Limitations of Existing Molecular Concurrent Training Evidence

There are currently insufficient human data providing any evidence of acute molecular interference occurring following concurrent exercise to explain the attenuated hypertrophy and strength response following concurrent training. However, it is unclear whether this is a consequence of a lack of the proposed mechanisms operating in human skeletal muscle, and/or the limitations of existing evidence, which are briefly discussed below.

7.1 Relationship between Acute Exercise Responses and Chronic Training Adaptation

Most acute molecular concurrent training studies provide a brief ‘snapshot’ of the acute adaptive events occurring in close proximity to concurrent exercise bouts [25–30]. However, many questions remain with regards to the long-

term molecular regulation of skeletal muscle adaptation and interference seen with concurrent training. First, there are limited data indicating a direct coupling between the acute molecular responses to exercise and the long-term phenotypic adaptations associated with chronic exercise training [14]. It is therefore unclear whether the acute molecular responses following concurrent exercise provide a valid 'readout' of the adaptive phenotype and potential interference induced if training was repeated long term. Indeed, the efficacy of the molecular markers commonly used to gauge the anabolic response to exercise and nutritional stimuli has been questioned [126, 127]. Only two human studies performed to date have directly measured protein synthesis rates following acute concurrent exercise [25, 26], whilst others have instead utilized proxy markers of protein synthesis and/or degradation [24, 27–30]. However, importantly, a direct coupling between mTORC1 signaling and protein synthesis rates does not always exist in humans [126], and rates of protein synthesis can be saturated at approximately 30 % of the maximal phosphorylation of p70S6K in rodents [128]. Given these apparent discordances, any minor interference to anabolic signaling responses following concurrent exercise may not reflect any potential interference to protein synthesis, and subsequently chronic muscle hypertrophy. Studies extrapolating the anabolic response and potential acute interference effect from acute signaling responses alone must therefore be interpreted with caution. Nevertheless, p70S6K phosphorylation following resistance exercise correlates well ($r = 0.82\text{--}0.99$) with chronic hypertrophy in both rat [129] and human [130, 131] models, supporting this as an appropriate proxy for chronic hypertrophy and potential marker for interference with concurrent training. Regardless of the methods used to gauge the post-exercise anabolic response, most acute concurrent exercise studies have been characterized by poor temporal resolution. For example, most existing studies have examined acute molecular responses up to 4 h post-exercise, whereas mTORC1 signaling persists for up to 24 h post-exercise [132, 133]. These studies may have therefore missed any potential effects of concurrent exercise on mTORC1 signaling occurring later than 4 h post-exercise. Further work employing extended time-courses is required to determine whether mTORC1 signaling is altered by concurrent exercise during the later recovery period.

7.2 Effect of Training Status on Acute Molecular Responses to Exercise

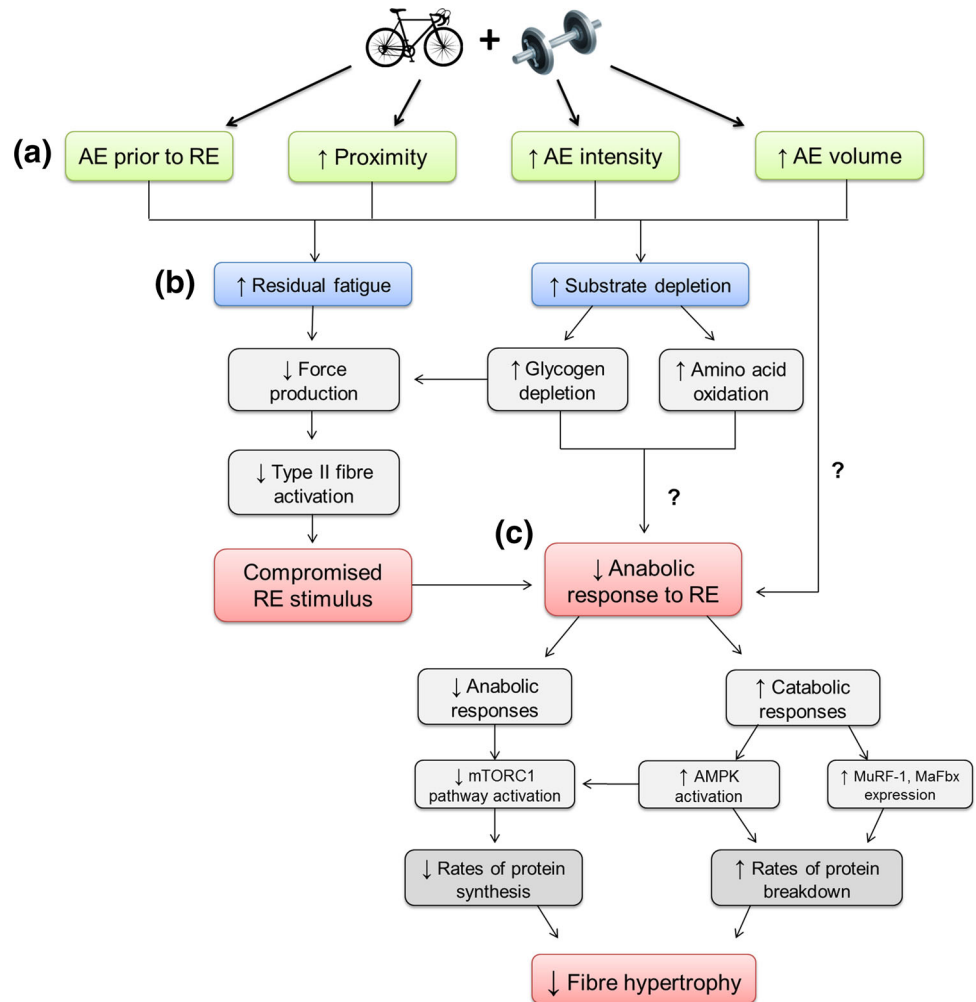
It appears likely that long-term concurrent training would modulate acute exercise responses (and potentially acute interference) over time, similar to that seen with single-mode exercise [59–63, 77, 134], whereby the acute

molecular profile to unaccustomed exercise bouts may represent a generalized, unrefined adaptive response [3, 62]. For example, while there is evidence of little divergences in mTORC1 signaling responses between resistance and endurance exercise in relatively untrained subjects [58, 61, 62], the mTORC1 pathway is indeed preferentially activated by resistance, but not endurance exercise, in training-accustomed individuals [61]. Additionally, while resistance exercise performed in the untrained state elicits comparable increases in both myofibrillar and mitochondrial protein synthesis rates, these responses become more refined following training, whereby only myofibrillar protein synthesis rates are increased in response to resistance exercise [62]. Evidence from chronically strength- or endurance-trained athletes [59, 60] also supports the notion that muscle phenotype, rather than the mode of exercise per se, can influence the molecular responses to divergent exercise modes. In this case, current acute concurrent training evidence in relatively untrained subjects should be interpreted with caution. Indeed, while most acute molecular concurrent training studies have utilized participants who were recreationally undertaking both resistance and endurance training [25, 27–29], some have used sedentary participants [26, 30] or participants not accustomed to both modalities [29]. Observations that early concurrent exercise bouts promote cumulative effects on protein synthesis and/or mitochondrial biogenesis signaling [26, 29, 30] may therefore be more reflective of the unfamiliarity to the exercise bout [53] rather than suggesting enhanced potential for chronic adaptation. Moreover, often overlooked is that the original study by Hickson [10] showed no detectable interference effect until the 8th week of concurrent training, suggesting that interference may not manifest until a certain training status is attained. Taken together, these results suggest that participant training status is an independent influence on exercise-induced molecular responses, and potentially the interference effect, and must be taken into consideration when interpreting existing concurrent training evidence. Future work examining the existence of molecular interference should employ participants who are accustomed to both exercise modes to account for the potentially confounding effects of training status on acute post-exercise molecular responses to exercise [59, 60]. Moreover, this further exemplifies the need for chronic (>8 weeks) training studies examining the potential modulation of acute interference following long-term concurrent training.

7.3 Effect of Nutrient Availability on Acute Molecular Responses to Exercise

It has become increasingly clear that nutrient availability exerts a profound effect on the adaptive responses to

Fig. 2 Conceptual framework for the potential role of individual concurrent training variables **a** in exacerbating the interference effect, either by **b** compromising the resistance exercise stimulus itself via increasing residual fatigue and/or substrate depletion, or **c** by attenuating the anabolic response to resistance exercise, subsequently limiting muscle fiber hypertrophy. *AE* aerobic exercise, *AMPK* adenosine monophosphate-activated protein kinase, *eEF2K* eukaryotic elongation factor 2 kinase, *MaFbx* muscle-atrophy f-box (atrogin 1), *mTORC1* mammalian (mechanistic) target of rapamycin complex 1, *MuRF-1* muscle ring-finger 1, *RE* resistance exercise, ↑ indicates increased/greater, ↓ indicates decreased/less



exercise training in human skeletal muscle [135, 136]. For example, the availability of muscle glycogen has been reported to modulate early molecular responses to both endurance and resistance exercise in a divergent manner [82, 84, 137]. Low carbohydrate availability appears to augment early signaling responses governing metabolic adaptation and mitochondrial biogenesis [84, 137, 138], whilst low muscle glycogen may compromise anabolic responses to resistance exercise [82]. However, the latter effect was recently questioned by a study showing no effect of muscle glycogen depletion on anabolic responses to resistance exercise [139]. Low muscle glycogen is associated with fatigue development [140, 141] and increased AMPK activity [142], which might inhibit anabolic responses induced by resistance exercise [82, 92]. It is also well accepted that essential amino acids can independently stimulate mTORC1 activation and subsequently increase protein synthesis rates [73, 143, 144] via interaction with the Rag GTP-ases (guanosine triphosphate-ases) [74]. Nutrient availability is therefore a potent modulator of acute molecular responses to exercise and skeletal muscle

adaptations following chronic exercise training [135] and needs to be considered when interpreting the concurrent training literature.

Most existing molecular concurrent training studies have employed designs whereby participants performed exercise in the fasted state [24, 25, 27, 28, 30], or were not provided with nutrients upon cessation of exercise [29], presumably to control for the independent effects of nutrient availability on acute molecular responses within skeletal muscle [135]. Performing exercise in the fasted state undoubtedly presents a heightened metabolic challenge within the muscle milieu, presumably increasing energy-sensing kinase activity (e.g., AMPK and eEF2K) with the capacity to suppress protein synthesis [23, 92], and promote autophagy [145]. The ability of amino acid ingestion to independently stimulate activation of anabolic signaling responses [73, 143, 144] suggests adequate nutrient availability may be essential for attenuating the potentially negative impact of endurance exercise and the associated molecular responses on protein synthesis [146]. Indeed, it is well established that ingestion of sufficient

protein in the early recovery period following resistance exercise is required to maximize muscle protein synthesis and subsequently muscle hypertrophy [135, 147]. Further, as muscle hypertrophy is an energetically demanding process, a positive energy balance may be required to support increases in muscle mass [148], and endurance exercise almost certainly interferes with this balance via continual substrate depletion and/or amino acid oxidation [149]. The potentially confounding effects of altering nutrient availability on molecular responses to exercise should therefore be considered when interpreting the concurrent training literature. Additionally, further work is required to fully elucidate the importance of nutrient availability on maximizing concurrent training adaptation.

8 The Role of Concurrent Training Variables in the Interference Effect

Given the multitude of potential concurrent training variables, the roles of many of these variables in acute and chronic interference remain incompletely resolved. Potentially, specific concurrent training variables might exacerbate molecular interference, either indirectly by compromising the 'quality' of the resistance exercise stimulus itself (e.g., via residual fatigue or substrate depletion), or directly by increasing the activity of proteins acting to inhibit protein synthesis and/or stimulate protein breakdown (see Fig. 2). Improving knowledge of the contribution of these variables to the interference effect is therefore critical to inform their prescription for maximizing the simultaneous development of muscle mass, strength, and endurance. Existing evidence has nevertheless begun to shed light on the potential roles of specific training variables in the concurrent interference effect.

8.1 Within-Session Exercise Order

A common and time-efficient concurrent training approach is to perform divergent exercise bouts together within a single exercise session. The order in which these exercise modes are performed may potentially modulate interference. However, minimal work has been done to ascertain whether an order-effect-dependent interference effect exists when concurrently training [150–155]. From a first-principles perspective, it would appear that if substrate depletion and/or residual fatigue [11, 156] from prior endurance training bouts compromises performance during subsequent resistance exercise, then undertaking resistance exercise prior to endurance exercise may alleviate these negative residual effects. However, given that metabolic signaling responses to endurance exercise that inhibit protein synthesis (e.g., AMPK activation) are generally

relatively transient (<3 h) [157, 158] compared with anabolic responses (i.e., mTOR and p70S6K phosphorylation) to resistance exercise (>24 h) [132, 133], performing resistance exercise after endurance exercise may allow these anabolic responses to proceed unimpeded during early recovery. Nevertheless, current performance-based evidence suggests that the effect of within-session exercise order on chronic interference may be limited [150–152], although performing resistance exercise prior to endurance exercise appears to augment neuromuscular and cardiorespiratory adaptations in the elderly [153, 155].

Two studies [27, 28] have addressed this question by examining acute signaling responses and gene expression after consecutive resistance and endurance training sessions completed in an alternating order (i.e., resistance prior to endurance exercise, and vice versa). The initial study [28] employed consecutive leg extension exercise (8 × 5 repetitions at 80 % 1RM) and continuous cycling (30 min at 70 % $\dot{V}O_{2peak}$) and noted a greater increase in MuRF-1 mRNA when cycling followed resistance exercise, while the reverse order caused increased phosphorylation of Akt and decreased IGF-1 mRNA expression [28]. No significant order effect was noted for TSC-2/mTOR/p70S6K activation and for the modest increases in PGC-1 α mRNA. Taken together, the results provided little evidence for any order-effect-dependent molecular interference, whilst the authors suggested that concurrent exercise did not promote optimal acute signaling responses associated with each exercise mode [28]. However, the lack of a single-exercise mode condition within this design [28] makes it difficult to speculate on the magnitude of any potential interference relative to resistance exercise alone.

A subsequent study from the same group [27] incorporated a resistance exercise session identical to the first [28], but combined this with a repeated-sprint cycling protocol (10 × 6-s sprints) performed either before or after the resistance exercise. These workers reported that concurrent repeated-sprint and resistance exercise promotes acute interference by attenuating translation initiation signaling (i.e., p70S6K and its downstream target, rps6 [ribosomal protein s6]), and increasing the mRNA abundance of mediators of protein degradation (i.e., MuRF-1). Divergent p70S6K phosphorylation responses were noted between exercise modes and with alternate exercise orders. Specifically, initial resistance exercise promoted increased p70S6K activation, whilst this response was attenuated when resistance exercise was performed after repeated sprints [27]. These authors concluded that performing repeated-sprint exercise in close proximity to resistance exercise attenuated anabolic signaling and increased catabolic activity, which likely represents acute interference of pathways governing resistance training-related adaptation [27]. Consequently, it was recommended that both exercise

modes be performed with a significant intervening recovery period to minimize acute interference, and that resistance training precede repeated sprints if performed within the same session.

8.2 Between-Mode Recovery Length (Proximity)

Given that performing concurrent exercise bouts in close proximity may represent a sub-optimal training scenario [27, 28], the recovery length allowed between undertaking concurrent exercise sessions is another important practical consideration. Potentially, residual fatigue and/or substrate (i.e., muscle glycogen) depletion from endurance training bouts may impact negatively upon force/power production [11, 156] and anabolic signaling responses [82] to subsequent resistance exercise, respectively. For example, following a bout of endurance exercise, force production of the exercised musculature is reduced for at least 6 h [43, 159–161], returning to baseline by 24 h post-exercise [160]. Compromised force/power production during resistance exercise would theoretically limit activation of higher-threshold (i.e., type IIx/IIb) motor units and fibers [162, 163], known to be most responsive to hypertrophy [7]. Indeed, the phosphorylation of p70S6K [70, 164] and mTORC1 [165] after resistance exercise is more pronounced in type II than in type I muscle fibers. Residual fatigue from prior endurance exercise also reduces the volume of work performed during subsequent resistance exercise [161], presumably limiting the potential for muscle hypertrophy. Finally, the transient activation of various proteins by endurance exercise that inhibit protein synthesis (e.g., AMPK and eEF2K) [92, 166] and mediate protein degradation (e.g., MaFbx and MuRF-1) suggests that commencing resistance exercise in closer proximity to endurance exercise may further compromise the anabolic response to resistance exercise. Allowing adequate recovery between concurrent exercise sessions may therefore attenuate any negative residual effects from endurance exercise on subsequent training bouts, consequently alleviating any interference. This approach also provides opportunity for carbohydrate and/or amino acid ingestion, essential to replenish muscle glycogen stores [167] and counteract the potentially detrimental impact of endurance exercise and associated molecular responses on protein synthesis [23, 92, 168], respectively.

The concept of between-mode recovery has been significantly under-researched in the concurrent training literature [12, 169]. In their meta-analysis of concurrent training literature, Wilson and colleagues [12] noted a trend (non-statistically significant) towards greater hypertrophy gains in concurrent training studies when resistance and endurance training were performed on separate days compared with on the same day. Most existing molecular

concurrent training studies have employed designs whereby concurrent resistance and endurance training bouts are performed in succession following only a brief recovery (e.g., 15–30 min) [25–28, 30]. These studies therefore provide little mechanistic insight into the effect of longer between-mode recoveries on both force/power production and anabolic responses to subsequent resistance exercise bouts. Work by Lundberg et al. [29] examined acute molecular responses to resistance exercise (2×7 bilateral leg press and leg extension repetitions) performed 6 h after aerobic exercise (40 min of continuous unilateral cycling at 70 % peak power output [$W_{\max}$]) compared with that seen after resistance exercise alone. These authors noted that divergent exercise modes performed after a significant intervening recovery period in the fed state resulted in elevated anabolic signaling (i.e., increased mTORC1 and p70S6K phosphorylation) and lowered myostatin gene expression compared with resistance exercise alone. Moreover, the addition of resistance exercise appeared to accentuate early PGC-1 α mRNA abundance, whilst prior endurance exercise did not compromise force and power production during resistance exercise. It was therefore concluded that divergent exercise modes can be successfully performed on the same day without compromising performance or the molecular responses mediating protein synthesis and mitochondrial biogenesis [29]. However, whether shorter recovery lengths would have exacerbated any putative molecular interference is unclear. Further work is therefore required to determine the role of between-mode recovery in concurrent interference, in addition to recovery strategies that may be employed during this period. Such information may help to develop practical training recommendations for the structuring of concurrent resistance and endurance exercise sessions to support maximal simultaneous development of resistance and endurance adaptation.

8.3 Endurance Training Intensity

Another important practical consideration is the intensity of endurance training employed in a concurrent training regimen. Recently, high-intensity interval training (HIT) has emerged as a potent exercise strategy for inducing signaling related to mitochondrial biogenesis (i.e., PGC-1 α expression) and associated health benefits (e.g., improved insulin sensitivity) typically associated with longer-duration, lower-intensity endurance exercise [138, 170–176]. Therefore, HIT represents a time-efficient strategy for promoting mitochondrial biogenesis and associated improvements in oxidative capacity and metabolic health. This high-intensity approach is also favored in conditioning programs tailored for enhancing anaerobic capacity and/or repeated sprint ability [37].

Despite the relevance of HIT for health and performance outcomes [172], little is currently known regarding the effects of incorporating HIT in a concurrent training regime on acute molecular interference and subsequent chronic adaptation to long-term concurrent training. Indeed, studies independently examining the effect of endurance exercise intensity on concurrent interference are scarce [177]. One study suggested no role for endurance exercise intensity on interference in physically active females [177]; however, training volume and frequency were comparably low and may have limited any potential interference effect seen with higher training volumes [12, 178, 179]. Most existing molecular research has employed low-to-moderate intensity endurance exercise protocols (e.g., 30–60 min at 65–70 % $\text{VO}_{2\text{max}}$, 40–90 min at 55–70 % W_{max}) [24–26, 28–30], whilst none have directly examined the effect of endurance exercise intensity on acute molecular and chronic performance interference. As aforementioned, Coffey and colleagues [27] observed that repeated sprints attenuated the anabolic response when performed concurrently with resistance exercise, and seemingly more so than in a previous study employing moderate-intensity continuous cycling [28]. However, little is known regarding the role of more practical HIT models involving longer work intervals interspersed with periods of active or passive recovery [170, 175, 180] on acute interference and subsequent chronic adaptation when performed concurrently with resistance training.

From a molecular perspective, higher-intensity endurance exercise (i.e., HIT) may exacerbate acute molecular interference when compared with lower-intensity (i.e., continuous) endurance exercise. For example, the activity of selected negative regulators of protein synthesis, such as AMPK and 4E-BP1, is increased by endurance exercise in an intensity-dependent manner [181, 182]. Moreover, the AMPK- $\alpha 1$ catalytic isoform, which purportedly plays a selective role in mTORC1 inhibition [94, 96], may be preferentially activated by higher [182], but not lower [183], endurance exercise intensities. Higher-intensity endurance exercise also appears to inhibit subsequent force production [43, 159], whilst lower-intensity continuous exercise may cause less residual fatigue [184]. Finally, higher exercise intensities are associated with increased glycogen depletion occurring predominantly in type II muscle fibers [185], which may exacerbate residual fatigue [140] and increase inhibitory AMPK activity [142]. Whether the capacity of higher-intensity exercise to cause greater metabolic perturbation in type II muscle fibers [185, 186] plays any role in potentially blunting anabolic responses within these fibers following subsequent resistance exercise remains to be determined. Regardless, given the potency of HIT for inducing favorable adaptations in skeletal muscle [187], along with the importance of HIT

for athletic performance [37], more work is required to delineate the role of endurance exercise intensity in the concurrent interference effect.

8.4 Endurance Training Volume

The possibility also exists that the total volume of endurance exercise, rather than the intensity per se, may be more crucial in mediating concurrent interference [12, 178, 179]. A role for volume-dependent interference is supported by studies reporting no interference with smaller endurance training frequencies (≤ 2 sessions per week) [40, 42, 188], whilst others have observed attenuated maximal strength with larger endurance training volumes (≥ 3 sessions per week) [10, 38, 41, 44, 179]. Greater attenuation of strength and hypertrophy (estimated via limb girth) has been shown to occur with greater frequencies of concurrent endurance exercise (3 days per week for each mode) than when endurance exercise was performed once per week [179]. Nevertheless, it remains to be determined whether the total weekly endurance training volume, or the training frequency per se, is the more critical factor mediating concurrent interference. If endurance exercise volume is key, low-volume HIT protocols [171, 180] might confer benefit when incorporated into a concurrent training regimen by limiting any potential volume-dependent interference effect, whilst also offering similar metabolic and performance benefits to traditional endurance exercise [171, 172]. Further work in this area is required to inform the manipulation of concurrent endurance training volumes and/or intensities in order to minimize their potentially negative impact on resistance training adaptations.

8.5 Endurance Training Modality

The endurance training modality employed in a concurrent training regime may also modulate interference following long-term concurrent training [11, 12]. Interestingly, the majority of concurrent training studies reporting an interference effect have incorporated running, and less often cycling, as the endurance training modality [11, 12]. It remains unclear what might account for any mode-specific interference effect, although it has been suggested that this may relate to the similarity between cycling and many strength outcome measures [12, 189], and/or the greater eccentric muscle damage induced by running compared with cycling [12]. Whether running exercise has the capacity to induce greater catabolic molecular activity and/or exacerbate residual neuromuscular fatigue in contrast to cycling, which may in turn exacerbate interference, is currently unclear. Relatively little is known regarding the impact of running exercise on acute post-exercise adaptive responses in skeletal muscle compared with cycling.

Indeed, no studies performed to date have examined the molecular responses to concurrent exercise incorporating running as the endurance training modality. Given that the majority of team-sport athletes (e.g., Australian football, soccer, rugby, etc.) employ running as the predominant endurance training modality and the anecdotal popularity of running in recreational concurrent training regimes, further work is required to examine the potential consequences of the endurance exercise modality on acute molecular interference and subsequent long-term adaptation to concurrent training.

9 Conclusion

While considerable performance-based evidence exists for the concurrent interference effect, limited data are available regarding the molecular factors responsible and the role of specific training variables in this phenomenon. The majority of current molecular data suggests that endurance exercise does not compromise early anabolic responses to resistance exercise, consequently providing little mechanistic insight into the interference effect seen following long-term concurrent training. However, findings from existing research are complicated by the multitude of potential concurrent training variables and the numerous independent factors capable of influencing acute molecular responses to exercise in human skeletal muscle. Thus, there is substantial difficulty in deducing practical training recommendations from existing research for minimizing interference between concurrent resistance and endurance exercise. Moreover, whilst considerable advances have been made with regards to our understanding of the molecular factors mediating training adaptation in skeletal muscle, these complex processes are incompletely resolved. Observations that anabolic signalling responses to exercise are not always directly coupled to protein synthesis, and that these acute responses are not necessarily predictive of chronic training adaptations, represent significant limitations to acute-based concurrent training studies. The possibility remains, therefore, that solely utilising early molecular responses to concurrent exercise to extrapolate any putative chronic interference effect may provide limited insight into factors mediating interference following long-term concurrent training. However, further elucidation of the molecular factors mediating the specificity of training adaptation in human skeletal muscle is warranted, which in turn may provide additional mechanistic insight into the concurrent interference phenomenon. Ultimately, improved understanding of the roles of individual concurrent training variables, including within-session exercise order, length of between-mode recovery, and endurance training volume, intensity and modality, in modulating the interference effect

is required to guide exercise prescription for simultaneously maximising divergent training adaptations. Future work should aim to further clarify the roles of these training variables in acute and particularly chronic interference in trained individuals to inform practical recommendations for minimising interference between concurrent resistance and endurance exercise.

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